

A Simple yet Useful Spiral Visualization of Large Graphs

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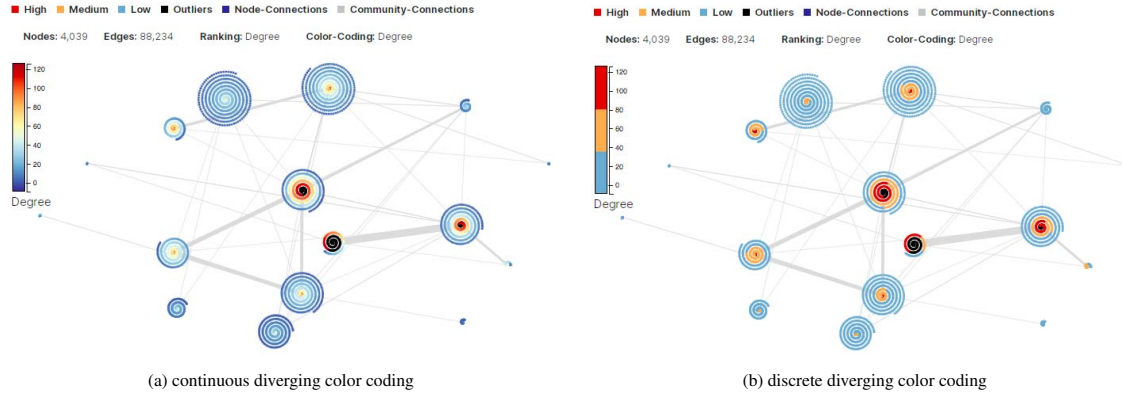


Figure 1: This figure presents two *Spiral Visualizations* on the *Facebook* dataset having 4,039 nodes and 88,234 edges. The Spiral Visualization on the left (a) uses continuous diverging color coding, while the one on the right (b) uses discrete diverging color coding to color the nodes. Each spiral in a spiral visualization represents a community within the dataset. The radius of the spiral represents the size of the community. Each edge represents the existence of inter-community edges between two communities, and the edge width represents the number of interconnections between them. Nodes are ordered and colored based on degree centrality to visualize and compare degree distribution in spirals.

ABSTRACT

We present a *Spiral Visualization* that facilitates users to visually comprehend large graphs. Spiral Visualization is a representation that highlights key aspects of networks, including the number, size, and density of communities, important or central nodes within communities, centrality distribution within communities, connections between communities, and connections between nodes. To facilitate analysis and comprehension of networks using various interaction techniques, such as zooming, tooltip, and highlight, we have implemented a Spiral Visualization dashboard. We conducted a qualitative user study incorporating observation, think-aloud protocols, and participant rating of confidence and easiness to assess the usability and suitability of our visualization. The findings suggest that our visualization is appropriate for the network tasks evaluated. However, tasks requiring color comparison, such as identifying the densest community and comparing community densities were found to be more challenging to perform.

Index Terms: Human-centered computing—Visualization—Visualization techniques—Graph Drawing; Human-centered computing—Visualization—Visualization design and evaluation methods

1 INTRODUCTION

Networks are characterized by communities [10], which are subsets of nodes that are densely interconnected among themselves compared to the rest of the network. It is challenging to visualize topological structures and communities in large networks that contain hundreds to thousands of nodes [22]. An important aspect of

analyzing networks is identifying central, influential, or important nodes in communities, which has applications in various fields such as marketing, social influence, network security, and epidemiology. For example, (1) identifying nodes that control virus propagation in an immunization network is essential for preventing epidemics, and (2) identifying important stations in transportation networks is critical for reliable transportation. While the *centrality measures* [5] quantify the importance of individual nodes within a network, they may not provide a comprehensive understanding of a node's centrality in relation to its connections within and across communities. *Our visual analytics solution provides an intuitive and flexible approach to data exploration that allows users to identify patterns and relationships between nodes and communities for analyzing node centrality.*

The Spiral Visualization allows users to explore multiple aspects of a network, such as the number of communities, the size of a community (the number of nodes in a community), edge-density [28] in a community (the ratio of links present in a community to the total number of possible links between community nodes), node connections (intra-community and inter-community connections), community connections, node centrality, and the distribution of node centrality values in communities. A user study was conducted with ten participants, and the results suggest that our visualization is suitable for most of the tasks evaluated. We used sixteen tasks based on these graph characteristics to assess the effectiveness of the Spiral Visualization.

2 RELATED WORK

Information visualization and graph drawing have gained much attention in recent years [9, 16, 33]. Node-link diagrams are commonly used for displaying sparse graphs, but become unreadable for dense graphs [12]. Adjacency matrix representations are ideal for dense graphs and community analysis [1, 14] but lack intuitiveness. The adjacency matrix represents degree centrality through colored cells in a

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row (or column). However, it is not suitable for visualizing other centrality measures because ordering nodes based on centrality in rows and columns hinders the perception of structural properties in the graph. Hybrid visualization [29] techniques, such as Nodetrix [15] and ChordLink [2], can integrate different drawing styles. Nodetrix represents dense subgraphs or intra-community connections using an adjacency matrix in a node-link diagram. Nodetrix can visualize central nodes based on degree and betweenness. ChordLink uses chord diagrams to represent dense subgraphs (or communities) in a node-link diagram that shows the global network structure. However, Nodetrix and ChordLink cannot be used to visualize large graphs and do not support other centrality measures like closeness and eigenvector centrality. Community-based aggregation [32] solutions can visualize large graphs, using super nodes to represent one or more nodes. Group-in-a-Box (GIB) [8] and ModuleGraph visualization [23] are suitable for representing large graphs. GIB assigns spatial regions to groups or communities to visually organize complex networks. ModuleGraph employs various patterns to illustrate interactions within each module (group of nodes), emphasizing the understanding of both intra-module and inter-module connections. Although some approaches [8, 24], utilize visual channels like node size and color saturation to convey node centrality, these channels may not be effective for representing quantitative values as small changes in size or color can be difficult to perceive.

Centrality is a measure of a node's importance in a network, and various centrality measures exist, such as *Degree Centrality* [25], *Closeness Centrality* [26], *Betweenness Centrality* [6], and *Eigen Centrality* [30]. Despite the importance of node centrality, it has not been given much attention in graph drawing literature [13, 18, 33]. The Spiral Visualization is specifically designed to visualize node centrality in different communities.

3 DESIGN OF SPIRAL VISUALIZATION

Our goal is to develop a visual solution for networks that satisfies three requirements: **(R1)** a compact and space-efficient representation for large graphs, **(R2)** positioning nodes belonging to a community close to each other for easy perception of communities, and **(R3)** visualizing node centrality through the most effective visual channel. As position is the most effective visual channel for perceiving quantitative data, our objective is to utilize node position to visually encode node centrality information in our visualization. We considered different shapes for representing nodes belonging to a community (e.g. straight lines, squares, and spirals) and concluded that arranging nodes in a spiral shape is a suitable choice due to the following reasons. A spiral is a path that curves around a fixed center point while continuously moving away from it. In a spiral layout, starting from the center, each successive node or point is positioned at a continuous increasing distance from the center. By utilizing the increasing distance, the spiral provides an effective method of displaying ordered data [7, 27, 35] using position as a visual channel **(R3)**. Next, spirals can display a large number of data points (or nodes) in a compact space **(R1)**. Finally, the close proximity of nodes in a spiral makes it an effective way to represent communities in a graph **(R2)**. Also, the spiral arrangement aids in reducing visual clutter by minimizing the overlap of nodes as they are positioned one after another in a spiral shape. Our approach is to order the nodes of a community based on their centrality and position them along the Spiral. The node with the highest centrality is placed in the center of the Spiral, followed by nodes with decreasing centralities.

4 METHOD

Undirected and unweighted graph data is taken as input, and a community detection algorithm (Louvain algorithm [4]) is used to group the graph nodes into clusters or communities. The community detection algorithm can reveal communities in the network, such as countries in an airline network or research topics in a citation

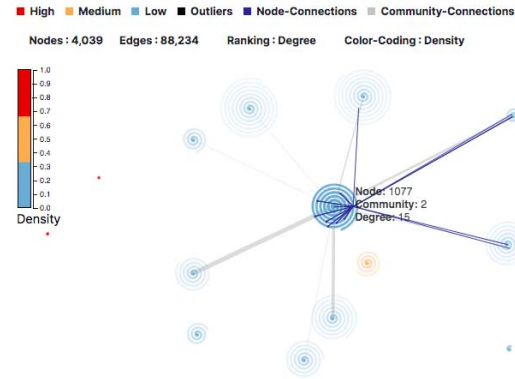


Figure 2: The figure demonstrates that when the user hovers the mouse over node-1077, its intra-community and inter-community connections are highlighted in dark blue color. Additionally, the community connections of the community to which node-1077 belongs are shown in gray color.

network. Next, each community is represented by a spiral (see Fig. 1). To obtain a Spiral, we order the nodes in a community based on their centrality [5] in descending order. The centrality is calculated on the whole network. Our dashboard allows users to interactively select the centrality measure such as *Degree Centrality* [25], *Closeness Centrality* [26], *Betweenness Centrality* [6], and *Eigen Centrality* [30]. The degree is the most widely used centrality measure (default setting). The ordered nodes are then placed along the spiral one after the other starting from the center of the Spiral. The centermost node in the spiral represents the most important node in the community with the highest centrality value, followed by nodes with decreasing centrality. Next, we draw edges in the Spiral Visualization. The Spiral Visualization has two types of edges showing *community connections* and *node connections*. A community connection is obtained by merging all the inter-community edges between two communities into a single edge, connecting the centers of the respective two Spirals. Combining multiple edges improves the readability of graph visualization and gives the intuition of the number of interconnections between two communities. The greater the number of inter-community node connections between two communities, the wider the width of the edge between them. Unlike community connections, node connections are shown only on demand to reduce visual clutter due to edge crossings. When the user hovers over a node of interest, all the edges to its adjacent nodes (within the same spiral or in different Spirals) are drawn using dark blue color as shown in Fig. 2.

4.1 Placement of Spirals

In this section, we provide details on how the placement of different spirals is determined. First, the graph nodes are partitioned into communities using Louvain algorithm [4], after which a coarse graph [3, 11, 17] is constructed. Coarse graph is a compressed version of the original graph that retains important properties such as communities and their connections. To build the coarse graph, nodes belonging to the same community are merged into a single super-node. Super-nodes S_i and S_j represent communities c_i and c_j , respectively, and there is an edge E_{ij} between S_i and S_j if and only if there is an edge between a subset of nodes in c_i and a subset of nodes in c_j . The weight of edge E_{ij} (w_{ij}) is the number of all the edges between c_i and c_j . Finally, we used a force-directed layout algorithm [19] on coarse graph for placing spirals in the visualization.

4.2 Color coding in Spiral Visualization

Users have the flexibility to select various color codings for different perceptions of graph data, with the default option being density-based color coding to visualize the edge density in communities. Our dashboard allows users to choose centrality-based color coding, which facilitates the visualization of centrality distribution in spirals. Users can choose between continuous or discrete diverging color schemes (as shown in Fig. 1). We have implemented a color-blind-friendly color scheme to ensure accessibility for all users. Nodes within the spiral are colored using blue, orange, and red colors to indicate low, medium, and high centrality values, respectively. Additionally, we use the color black to represent outliers, which correspond to nodes with very high centrality values. Instead of using color saturation, we opted for color hue to differentiate among low, medium, high, and very high centrality values. Degree centrality typically follows a power-law distribution in real-world graphs (most nodes have a low degree and very few have a very high degree). As a result, assigning divergent colors between the minimum and maximum degree centrality values can be challenging, as most nodes will fall into the low centrality category, resulting in them being colored the same. Outliers or nodes with very high centrality values must be identified, as they can distort visualization and hinder analysis. To identify outliers, we use the interquartile range (IQR) [31, 34], and any centrality value outside of IQR is colored black in Spiral Visualization. The quantitative centrality values within the Interquartile Range (IQR) are converted into categorical data by equally dividing the IQR into three distinct categories: low, medium, and high values. Each category is then assigned a unique color. The color coding based on degree centrality is demonstrated in Fig. 1b. In the Facebook dataset, the minimum and maximum degree centrality values are 1 and 1045, respectively, and the IQR is (1-126), any node with a degree greater than 126 is colored black.

4.3 Implementation of Spiral Visualization

The Spiral Visualization dashboard was developed using the d3.js JavaScript library. To demonstrate the basic concept (as shown in Fig. 1a) and to conduct a user study, we used the Facebook dataset (<https://snap.stanford.edu/data/ego-Facebook.html>). The dataset consists of 4,039 nodes, representing individuals, and 88,234 edges, representing friendship relationships between them. As part of the evaluation process (outlined in Sect. 5), we created a demonstration video using the LastFM dataset (<https://snap.stanford.edu/data/feather-lastfm-social.html>). LastFM is a music-focused social platform where users follow each other based on musical preferences. The dataset contains 7,624 nodes representing users and 27,806 edges representing mutual follower relationships between them. The source code and datasets for Spiral Visualization are available at <https://github.com/Garima17/SpiralVisualization>.

5 USER EVALUATION

Using a qualitative methods approach, we conducted a user study to assess the effectiveness of Spiral Visualization in comprehending graph characteristics. The study involved analyzing data from observations, think-aloud protocols, and participant ratings of confidence and easiness for each task performed. The study had three main objectives, testing the suitability of Spiral Visualization for graph analysis, identifying any usability issues during analytic tasks, and determining potential design improvements to enhance usability.

5.1 Participants and setting

In our user study, we asked the participants about their familiarity with graphs and visualizations. The participants who lacked knowledge in these areas were excluded, along with those below the age of 18 years. To ensure the data quality, we excluded the participants who did not pass the attention-check tests or completed the study in less than ten minutes. Ultimately, the study included ten

participants with engineering background, aged between 19 and 35. Among them, six were computer science engineering students, and the remaining four were employed in software companies.

We provided a brief five-minute introductory video to explain the basic concepts of Spiral Visualization <https://youtu.be/CQmDE57U5mg> and an six-minute demonstration video to illustrate how to navigate the dashboard <https://youtu.be/cvLdXATHIXY>. A Spiral Visualization was then shown on a 15" Retina Display. We used continuous diverging color coding as shown in Fig. 1a (left) for the user study. The participants were given basic instructions and examples before commencing each task, after which they were asked to perform tasks by interacting with the visualization through the mousepad. We used Google forms for our user study, on a separate tablet computer during the session. The participants could submit their answers directly through a tablet. A total of sixteen tasks (Table 1) were analyzed to determine how well Spiral Visualization facilitates users to comprehend graph characteristics stated in Sect. 1. These sixteen tasks were created by taking generic task taxonomies [21] as starting points. The participants were asked to rate each task based on their confidence in their answers and the extent to which they perceived the task as easy. We use a four-point Likert scale to rate each task's easiness and confidence level. The easiness scale ranges from 1 to 4 (1 = very easy, 2 = easy, 3 = difficult, 4 = very difficult). Similarly, the confidence scale ranges from 1 to 4 (1 = very confident, 2 = confident, 3 = unconfident, and 4 = very unconfident). Each session took approximately 45–60 min. The sessions were audio and screen-recorded using QuickTime Player. Our user study is based on the evaluation techniques [20] used in data visualization. To gather data, we utilized various methods including direct observations of participants while performing tasks, think-aloud protocols, screen captures, and participants' self-reported ratings of confidence and easiness for each task. A follow-up interview was conducted to obtain the participants' opinions regarding the usefulness and user satisfaction with the visualization. We analyzed the data collected from the qualitative study using an inductive thematic analysis approach. Through careful examination of the data, meaningful codes were generated and categorized into two key themes: challenges, and desired additional features. Each theme was analyzed individually. Finally, we presented the level of easiness, and confidence ratings estimated by the participants in Table 1 where important results are highlighted using green color. As this is a small exploratory study, the validity of results is limited and could further be investigated in a larger user study.

5.2 Results

Among the ten participants, nine could successfully identify the number of communities within the network (*Task 1*). These individuals expressed confidence in their solution and found "*Task 1*" easy with Spiral Visualization. However, one participant reported that counting the number of communities might be challenging if the spiral size is too small.

We have identified three tasks to evaluate the effectiveness of Spiral Visualization in understanding the size of communities. "*Task 2*" requires the user to identify the largest community, "*Task 3*" involves identifying the smallest community, and "*Task 4*" involves comparing the sizes of communities. One out of ten participants struggled to correctly identify the largest community and rated "*Task 2*" as difficult. Evaluation of *Task 2* conveys that if the communities or spirals have a similar number of nodes, participants may find it challenging to differentiate which community is larger. *Task 3* was not answered correctly by two participants. One participant found *Task 3* difficult and lacked confidence in their solution. This could be because identifying the smallest one among all the small spirals is difficult. Although all participants were able to compare

Table 1: The Table shows the number of participants (out of 10) and their levels of confidence and easiness while performing each task.

		Confidence				Easiness			
		1	2	3	4	1	2	3	4
	Number of communities in a network.								
Task 1	How many communities are present in the graph?	2	8	0	0	2	7	1	0
	Size of the community								
Task 2	Identify the largest community.	2	8	0	0	1	8	1	0
Task 3	Identify the smallest community.	3	6	1	0	3	6	1	0
Task 4	Compare the size of two communities.	8	2	0	0	7	2	1	0
	Density of the community								
Task 5	Identify the densest community.	3	4	3	0	3	4	3	0
Task 6	Identify low-density communities.	7	3	0	0	6	4	0	0
Task 7	Compare the density of two community.	5	5	0	0	4	5	1	0
	Node connections (a)								
Task 8	Can you visualize a node's inter-community and intra-community edges?	6	4	0	0	6	3	1	0
	Community connections (b)								
Task 9	Determine whether two communities are connected or not.	8	2	0	0	8	2	0	0
Task 10	Find the total number of communities adjacent to a community.	7	3	0	0	6	4	0	0
	Number of connections between two communities (c)								
Task 11	Identify which communities share the highest number of inter-community connections.	7	30	0	0	7	2	1	0
	Node centrality (d)								
Task 12	Find the most central node in a community based on the degree.	7	3	0	0	4	5	1	0
Task 13	Find the least central node in a community based on closeness.	5	4	1	0	4	4	1	1
Task 14	Can you compare the centralities of two nodes in a community?	7	3	0	0	7	3	0	0
	Distribution of node centrality values in communities.								
Task 15	Can you compare the centralities of two nodes in different communities	4	3	3	0	2	7	1	0
Task 16	Can you visualize the centrality distribution in communities?	5	5	0	0	5	4	1	0

the communities correctly, one participant reported it was a difficult task.

Next, we have three tasks to evaluate the suitability of Spiral Visualization in comprehending the density of communities. *Task 5* requires identifying the community with the highest density, while *Task 6* involves identifying all communities with low edge density. *Task 7* involves comparing the edge density of two communities. Out of 10 participants, 4 were unable to identify the densest community in the network, and 3 found this task challenging and lacked confidence. Additionally, 3 participants could not accurately compare the densities of two communities (*Task 7*). The error rate for Tasks 5 and 7 was high due to the color scheme used in Spiral Visualization, which made it difficult for participants to distinguish between darker and lighter shades of the same color. However, when the participants were asked to identify all low-density communities (*Task 6*), all participants were able to answer correctly, rating it as easy and showing high confidence. Hence, it can be inferred that colors can be utilized to estimate graph properties (e.g. *Task 6*), but it is challenging to determine precise values (e.g. *Task 5* and *Task 7*).

All participants were able to correctly complete tasks related to node connections, community connections, the strength of community connections, and node centrality, and they found the tasks to be easy. Additionally, all participants displayed confidence in their solutions. Therefore, the Spiral Visualization approach was effective in completing *Tasks 8 to 14* (a,b,c,d in Table 1).

Finally, to assess centrality distribution, we introduced two tasks, *Task 15* involved comparing node centralities across communities and *Task 16* involved visualizing centrality distributions. While all participants correctly answered *Task 15*, three of them lacked confidence, as the task involved comparing node colors and zooming. Additionally, two participants had difficulty visualizing the centrality distribution accurately (i.e., *Task 16*). It appears that more training is required to effectively visualize centrality distributions.

5.3 Observations

Four main challenges were identified during the qualitative evaluation and follow-up interviews: (i) **small communities**: visualizing

small communities (e.g., with only 20 nodes) can be challenging, and users may overlook them. Additionally, comparing their size and color can be difficult, making it difficult to analyze them effectively through Spiral Visualization, (ii) **interacting with central nodes**: nodes near the center of a spiral are densely packed, which makes it difficult to use interaction techniques (like a tooltip) on such nodes. Zooming in is necessary to interact with nodes near the spiral center effectively, (iii) **distinguish colors**: it can be difficult to differentiate precisely between communities with similar densities based on color saturation alone, and (iv) **identifying precise values**: Spiral Visualization can provide an estimate of large, small, dense, or low-density communities; however, it may be challenging to determine exact values, such as the largest or smallest community, or the densest or lowest-density community, using Spiral Visualization alone.

6 CONCLUSION

Our paper presents Spiral Visualization, a simple yet useful graph visualization approach that provides insights into various graph properties, such as the number and size of communities, community density, node centrality, and connectivity between communities and nodes. We present the results of a user study that assessed its suitability and usability. In the future, we plan to improve the Spiral Visualization by incorporating additional features suggested by users during the qualitative study. These features include displaying community labels on demand, providing a separate window for exploring the details of selected communities, adding a community-based search function, and filtering nodes based on their attributes. Additionally, we also intend to extend Spiral Visualization to handle directed and weighted graphs, thereby broadening the applicability of our technique and making it useful for a wider range of graph types and applications.

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